POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Compiler Design

Semester: Spring

Year : 2026
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

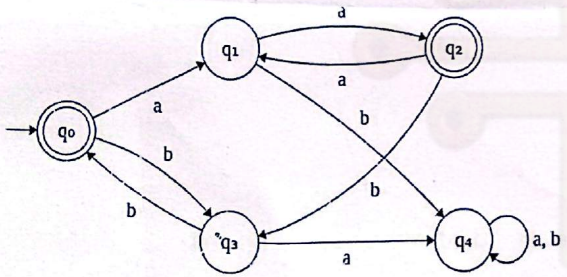
Attempt all the questions.

1. a) Imagine you are assigned as a role of compiler designer, explain about the different task/phase that you need to look in order to design a good compiler for any language. 7

   b) Explain the role of the scanner and describe how words (tokens) are recognized. Explain. 8

OR

Why state minimization in DFA is performed? Perform the DFA minimization for the given DFA:



2. a) Construct NFA for following Regular Expressions using Thompson's Construction. 8

   i. $(a + b)^*$

   ii. $((a^*| b)cd)^*$

   b) Convert the NFA obtained from a given regular expression (a|b)*abb into an equivalent DFA using the subset construction method. 7

3. a) What is LL(1) Grammar? Consider following grammar and compute FIRST and FOLLOW and also construct LL(1) parse table. 8

   S → aBDh
   B → cC
   C → bC | ϵ
   D → EF
   E → g | ϵ
   F → f | ϵ

   b) What is LR(1) Grammar? How do we construct LR(1) parse table? Consider following grammar and construct LR(1) parse table and parse string ***'abcba'*** by LR(1) parsing. 7

   S → aSa
   S →bSb
   S →c

OR

Construct the SLR parsing table for the given grammar.

E → E + T|T
T → TF|F
F → F*|a|b.

4. a) Explain in detail about different parser used in compiler. How the parsed grammar is moved to intermediate code generation explain with example. 8

   b) Explain graphical intermediate representation with syntax trees and graphs. Represent the expression x= a+a*(b-c)+(b-c)*d using three address code. 7

5. a) What is a symbol table? Explain how a hash table is used for building a symbol table. 8

   b) What are basic blocks and flow graphs? Explain with an example how optimization of basic blocks can be performed? 7

6. a) What is data-flow analysis? Explain the procedure for solving data-flow equations. 8

   b) Explain in detail about the machine dependent and machine independent techniques along with their example. 7

7. Write short notes on: **(Any two)** 2×5

   a) Interpreter vs. Compiler
   b) Register Allocation and Assignment
   c) Next Use Information

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